

## **Lecture 5 and beyond: Light as an EM Wave**

- **Derivation of wave equation from Maxwell's equations, in vacuum and in clear dielectric**
- **Special type of solution: Monochromatic plane waves of fixed polarization (express other solutions as superpositions of these)**
- **Maxwell's equations  $\rightarrow$  boundary conditions determining reflection and transmission at an interface (Fresnel relations)**

**\*\*today's material is covered in Steck Chapter 4\*\***

**Alert: There is a typo in Steck Equation 4.16**

# Traveling EM wave in vacuum:

*with no charges or currents*

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} \quad \text{0}$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \vec{\nabla} \times \vec{B} = \underbrace{\mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}}_{\text{0}}$$

$$\left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \vec{E} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\text{Wave traveling at speed } c_0 = \frac{1}{\sqrt{\mu_0 \epsilon_0}} .$$

# Maxwell's Equations in material:

*with no free charges or currents*

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

*Polarization density  $\vec{P}$*

$$\vec{\nabla} \cdot \vec{P} = -\rho_{\text{bound}}$$

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P}$$

$$\vec{\nabla} \cdot \vec{D} = \cancel{\rho_{\text{free}}}^0$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

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$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

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$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{j} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

*Magnetization density  $\vec{M}$*

$$\vec{\nabla} \times \vec{M} = \vec{j}_{\text{magnetization}}$$

$$\vec{H} = \frac{\vec{B}}{\mu_0} - \vec{M}$$

$$\vec{\nabla} \times \vec{H} = \cancel{\vec{j}_{\text{free}}}^0 + \frac{\partial \vec{D}}{\partial t}$$

# Maxwell's Equations in **linear media**:

*with no free charges or currents*

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

Polarization density  $\vec{P}$

$$\vec{\nabla} \cdot \vec{P} = -\rho_{\text{bound}} \quad \vec{P} = \chi_e \epsilon_0 \vec{E}$$

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P}$$

$$\vec{E} = \frac{1}{(1 + \chi_e)\epsilon_0} \vec{D} = \frac{1}{\kappa_e \epsilon_0} \vec{D} = \frac{1}{\epsilon} \vec{D}$$

$$\vec{\nabla} \cdot \vec{D} = \rho_{\text{free}} \quad \text{0}$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{j} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

Magnetization density  $\vec{M}$

$$\vec{M} = \chi_m \vec{H} \quad \vec{\nabla} \times \vec{M} = \vec{j}_{\text{magnetization}}$$

$$\vec{H} = \frac{\vec{B}}{\mu_0} - \vec{M}$$

$$\vec{B} = (1 + \chi_m) \mu_0 \vec{H} = \kappa_m \mu_0 \vec{H} = \mu \vec{H}$$

$$\vec{\nabla} \times \vec{H} = \vec{j}_{\text{free}} + \frac{\partial \vec{D}}{\partial t} \quad \text{0}$$

# Traveling EM wave in linear media:

$$\epsilon = \kappa_e \epsilon_0$$

$$\mu = \kappa_m \mu_0$$

$$\approx \mu_0$$

*for us starting  
next time*

$$\vec{\nabla} \cdot \vec{D} = \rho_{free}$$

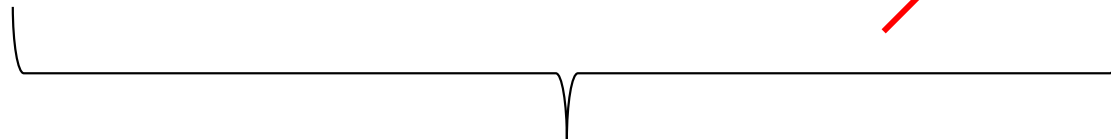
$$\vec{E} = \frac{1}{\kappa_e \epsilon_0} \vec{D} = \frac{1}{\epsilon} \vec{D}$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{B} = \kappa_m \mu_0 \vec{H} = \mu \vec{H}$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \times \vec{H} = \vec{J}_{free} + \frac{\partial \vec{D}}{\partial t}$$



$$\left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \vec{E} = \mu \epsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\text{Wave traveling at speed } c = \frac{1}{\sqrt{\mu \epsilon}} = \frac{1}{\sqrt{\kappa_m \kappa_e}} \frac{1}{\sqrt{\mu_0 \epsilon_0}} \equiv \frac{c_0}{n}.$$

*n is material's "index of refraction"*